

Predator Pressure and Habitat Complexity Determine Native Small Mammal Presence in Remnant Vegetation of South-Western Australia

Juan Seco a

a University of Western Australia

Summary

Small native mammals in south-western Australia are vulnerable to population decline due to habitat fragmentation, which disrupts vegetation structure and increases exposure to predators. While remnant vegetation patches in fragmented agricultural landscapes can support native fauna, their ability to sustain mammal biodiversity varies depending on the disturbance level, vegetation complexity, and predators' presence. To analyse these effects, this study assessed the richness and abundance of native small mammals across three remnant patches that differed in fencing, vegetation structure, and disturbance condition. Richness and abundance were highest in the least disturbed, fenced site with dense understorey and were completely absent in the most disturbed patch. Predator activity was significantly higher in disturbed areas, particularly in Elliott Corner, compared to Bush ($\chi^2 = 9.71$, $p = 0.0078$; Dunn's $p = 0.0095$), where native mammals were absent. Pearson correlations confirmed strong negative associations between red fox abundance and all native mammal species ($r = -0.72$ to -0.77), which supports the role of predators in limiting small mammal persistence. The principal components analysis revealed vegetation structure gradients aligned with mammal distribution, suggesting that understorey complexity contributes to habitat suitability. These findings collectively highlight the need to restore structural complexity and control invasive species to enhance native mammal persistence in fragmented landscapes.

Keywords: habitat fragmentation, understorey complexity, native small mammals, predator control, remnant vegetation

Implications for Managers

- Restoring and maintaining understorey complexity improves habitat quality and limits predator efficiency.
- Fencing and buffer zones help mitigate fragmentation effects in small or isolated patches.

- Coordinated control of red foxes and European rabbits reduces predation on native fauna.
- Landscape-scale planning should integrate vegetation restoration and invasive species management.

Introduction

Small native mammals in Australia are particularly affected by habitat fragmentation, which represents a critical driver of biodiversity loss, where extensive agricultural development and urban expansion have resulted in the isolation and degradation of remnant vegetation (Jones et al. 2021). This process modifies the spatial configuration of habitats by reducing patch size, increasing isolation, and exposing wildlife to edge effects, resource scarcity, and elevated predation risk (Holland & Bennett 2009). These impacts are particularly severe for small mammals, which are highly sensitive to habitat disturbance (Molloy & Davis 2017).

Native mammals such as the bush rat (*Rattus fuscipes*), brushtail possums, and small marsupials have faced significant pressures at the regional scale in south-western Australia. These include substantial declines in both distribution and abundance due to habitat fragmentation, introduced predators, and reduced structural complexity in vegetation (Molloy & Davis 2017). Nevertheless, remnant patches continue to play a crucial role by providing habitat refuges and movement corridors for numerous species (Fischer & Lindenmayer 2007).

This study explores the impact of remnant vegetation structure and associated stressors, such as predator presence and patch condition, on native small mammal richness and abundance in a fragmented agricultural landscape of south-western Western Australia. Identifying these relationships can guide conservation strategies that prioritise structural habitat attributes and predator management to support biodiversity in fragmented systems (Holland & Bennett 2009).

Methodology

The survey was conducted on a farming property near Narrikup, Western Australia. The property contains seven patches of remnant native bushland embedded within an agricultural matrix. On 7 March 2025, twenty Boly BG636-K trail cameras were deployed across three remnant vegetation sites: Bush, Twin Water, and Elliott Corner. Ten cameras were installed along two transects in the Bush site (five per transect), and five were placed in Twin Water

and five Elliott Corner. Cameras were mounted 40–50 cm above ground, oriented towards open areas to maximise detection probability, and programmed to record date, time, and GPS coordinates for each trigger event. All cameras were retrieved on 15 April 2025 (see Appendix 1 for camera placement details). During camera retrieval, vegetation structure was characterised at each site using a 10 × 10 m quadrat placed directly in front of the camera. Within each quadrat, visual estimates were made of percent cover for the following vegetation strata: trees 10–30 m, trees <10 m, shrubs >2 m, shrubs 1–2 m, shrubs <1 m, grasses, herbs, sedges, and succulents. Dominant plant species were also recorded for each stratum to describe habitat structure and composition at each camera location. Captured images were reviewed manually to identify vertebrate fauna. Species were identified using regional field guides and verified through online databases. For each detection, species identity, date, time, and camera location were recorded. Images triggered by non-target events (e.g. vegetation movement, livestock) were excluded from analysis. All statistical analyses were performed in R (version 4.5.0). Native small mammal richness, defined as the number of species detected per camera, was summarised using the mean and standard deviation per paddock. Boxplots and barplots were used to compare richness, abundance, and species composition across paddocks (Figure 1ba-f). Differences in richness, abundance, and predator activity were tested using Kruskal–Wallis tests with Dunn’s post hoc comparisons (Table 1). Pearson correlations assessed associations between predator abundance and native mammal presence (Figure 1g).

Results

Three native mammal species were recorded across the study area: Bush Rat (*Rattus fuscipes*), Common Brushtail Possum (*Trichosurus vulpecula*), and Gilbert’s Dunnart (*Sminthopsis gilberti*). Invasive mammals included the European Rabbit (*Oryctolagus cuniculus*). Predator species detected were Red Fox (*Vulpes vulpes*), Feral Cat (*Felis catus*), and Australian Raven (*Corvus coronoides*). The richness of native mammals varied among the three remnant sites, with Bush recording the highest values, followed by Twin Water, and no native species detected at Elliott Corner (Figure 1a). Predator richness showed an inverse pattern, with Elliott Corner and Twin Water recording higher values than Bush (Figure 1b). Total species abundance data indicated that Bush supported a higher abundance of native mammals compared to the other patches, with low counts of invasive mammals (European Rabbit) and predators (Figure 1c). In contrast, Elliott Corner showed higher abundances of invasive mammals and predators, with native mammals being absent (Figure 1d). Twin Water showed moderate values for all species groups (Figure 1e). The vegetation structure, as shown in mean percentage cover by strata,

varied across paddocks: Bush had a greater cover of low and mid-storey shrubs, Elliott Corner had elevated herb and tall tree cover. In contrast, Twin Water showed higher grass and lower shrub cover (Figure 1f). The Pearson correlation analysis showed strong negative relationships between Red Fox abundance and all native mammals (Bush Rat: $r = -0.721$; Gilbert's Dunnart: $r = -0.721$; Common Brushtail Possum: $r = -0.774$), and positive correlations with invasive species (European Rabbit: $r = 0.626$; Australian Raven: $r = 0.970$) (Figure 1g).

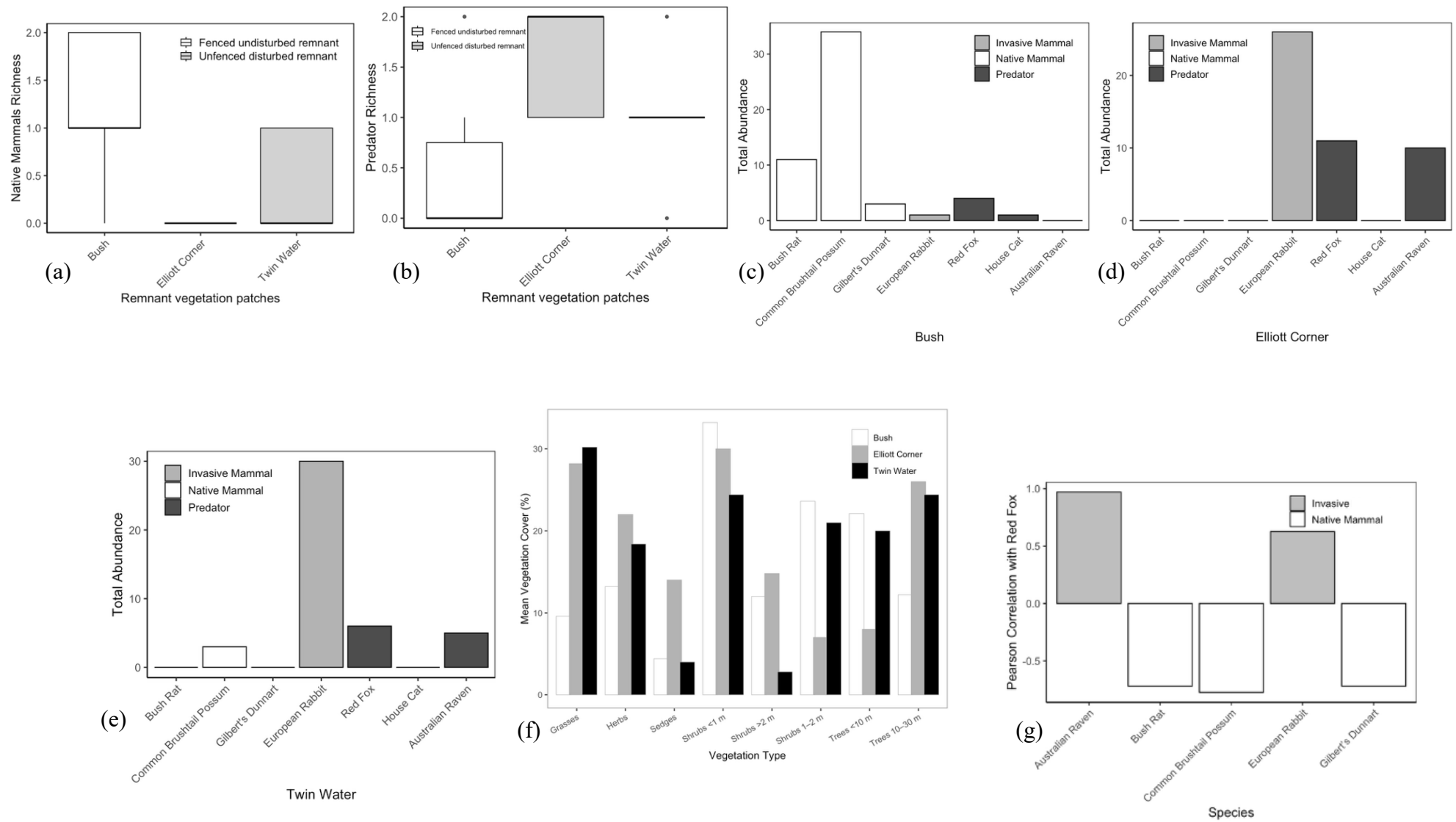


FIGURE 1. (a–b) Native mammal and predator richness by remnant type. (c–e) Abundance of native mammals, invasive mammals, and predators by species. (f) Mean vegetation cover by patch. (g) Pearson correlation coefficients between Red Fox abundance and small mammal species.

The Kruskal–Wallis test indicated that native mammal richness significantly differed among remnant vegetation patches. Post hoc comparisons showed a significantly higher richness in Bush compared to Elliott Corner, whereas the differences between Bush and Twin Water, as well as between Elliott Corner and Twin Water, were not significant. Predator richness also varied significantly between patches, with pairwise tests identifying a significant difference between Bush and Elliott Corner. For native mammal abundance, excluding rabbits, the Kruskal–Wallis test was likewise significant, with Dunn’s test indicating a difference between Bush and Elliott Corner, but not between Bush and Twin Water or Elliott Corner and Twin Water. Predator abundance showed a similar pattern, with a significant difference observed between Bush and Elliott Corner. In contrast, total mammal abundance, including rabbits, did not differ significantly (Table 1).

Table 1. Kruskal–Wallis test statistics and adjusted *p*-values from Dunn’s post hoc comparisons for small mammal response variables across paddocks.

Variable	Kruskal–Wallis Test Results		Dunn’s Post Hoc Pairwise Comparisons (Adjusted <i>p</i> -values)		
	Kruskal– Wallis χ^2 (df)	Kruskal– Wallis <i>p</i> -value	Bush vs Elliott Corner	Bush vs Twin Water	Elliott Corner vs Twin Water
Native Mammal Richness	8.71 (2)	0.0128	0.0142	0.1205	0.3514
Predator Richness	7.68 (2)	0.0215	0.0199	0.229	0.266
Native Mammal Abundance (excl. rabbits)	8.75 (2)	0.0126	0.0205	0.0637	0.5581
Predator Abundance	9.71 (2)	0.0078	0.0095	0.0814	0.3740
Total Mammal Abundance (incl. rabbits)	0.11 (2)	0.948	–	–	–

Note: Values are chi-squared statistics with associated degrees of freedom and *p*-values from Kruskal–Wallis tests, followed by adjusted *p*-values from Dunn’s pairwise comparisons (Benjamini–Hochberg correction). Values in bold indicate statistically significant differences ($p < 0.05$).

Discussion

This study aimed to assess whether the richness and abundance of native small mammals in remnant vegetation patches in southwestern Western Australia are influenced by vegetation

structure, predator presence, and fencing. As expected, the least disturbed patch of remnant vegetation supported higher native mammal richness and abundance and lower predator activity. Bush (fenced and undisturbed), Twin Water, and Elliott Corner (unfenced and disturbed) represent a gradient of habitat condition that allowed comparisons of how vegetation structure and disturbance influence native mammal communities. Differences in vegetation structure and predator presence across patches were associated with significant variation in native mammal richness and abundance, with the strongest statistical contrasts observed between Bush and Elliott Corner (Figure 1b, 1d, 1e, 1f) (Table 1).

Native mammals were only detected in the remnant with the greatest understorey complexity. Bush had the highest native mammal richness and abundance (Figure 1a, 1c), corresponding with its elevated cover of shrubs 1–2 m and >2 m, herbs, and sedges (Figure 1f). These vegetation strata likely provide key microhabitats and visual barriers that reduce predation risk, a pattern supported by previous findings highlighting the importance of dense understorey for the refuge and concealment of small mammals. (Short et al. 2011; Holland & Bennett 2009). In contrast, Elliott Corner, which lacked meaningful shrub structure and was dominated by herbaceous layers and tall trees (Figure 1f), showed a complete absence of native mammals (Figure 1a, 1d). This suggests that vertical simplification of vegetation limits available shelter and increases vulnerability to predation and environmental stress (Yeatman et al. 2016). Twin Water, although possessing some ground-layer vegetation, had limited shrub development and low native mammal detections (Figure 1e), indicating that sparse or low-lying cover alone may not suffice in sustaining viable populations. These site-level differences align with broader evidence from fragmented Australian landscapes, where structurally complex remnants support higher persistence of native fauna by maintaining habitat quality and reducing exposure to edge effects (Fischer & Lindenmayer 2007; Molloy & Davis 2017).

Red foxes were identified as key predators, showing strong negative associations with all native mammal species (Figure 1g), and their abundance was significantly higher in Elliott Corner compared to Bush (Table 1). This pattern likely reflects the combination of open vegetation and the absence of physical barriers, which together facilitate predator movement and access. In addition, European rabbits were more common in the same disturbed patches (Figure 1d, 1e, 1f), potentially serving as prey that sustains predator populations (Yeatman et al. 2016). Australian Ravens, frequently observed in open, grazed areas, may also intensify pressures through nest predation or scavenging (Sazima 2020). The simultaneous presence of foxes,

rabbits, and ravens in these disturbed patches reflects a reinforcing dynamic where multiple invasive species thrive under the same degraded structural conditions, collectively amplifying negative impacts on native fauna (Holland & Bennett 2009; Short et al. 2011). These overlapping interactions correspond with the observed declines in native mammal richness and abundance (Figure 1a–e; Table 1), supporting the view that structurally simplified remnants, which lack physical and vegetative barriers, are inadequate for sustaining native mammal communities (Fischer & Lindenmayer 2007; Moore et al. 2021).

Effective management of remnant vegetation should prioritise the restoration and maintenance of understorey complexity, which is critical for improving habitat quality and reducing predator efficiency. In small or isolated patches, strategies such as fencing, enhancing connectivity, and establishing buffer zones can help mitigate the impacts of fragmentation. Additionally, integrated control of invasive species is essential. Coordinated management of red foxes and European rabbits can reduce direct predation on native fauna and interrupt the ecological feedback loops that sustain predator pressure and contribute to vegetation degradation. These measures should be incorporated into landscape-scale conservation planning to support the long-term persistence of native mammal populations in fragmented systems (Holland & Bennett 2009; Jones et al. 2021; Molloy & Davis 2017).

The six-week sampling period, conducted in early to mid-autumn, did not capture seasonal variation in mammal or predator activity. To address this, future studies should include multi-seasonal monitoring for temporal dynamics. The small number of patches and lack of replication across remnant types also limit the generalisability of results; expanding sampling across a broader range of remnant conditions would strengthen inference. In Bush, the linear placement of camera traps may have undersampled habitat variability, suggesting that future designs could benefit from stratified or randomised placement to improve spatial representativeness.

Acknowledgments

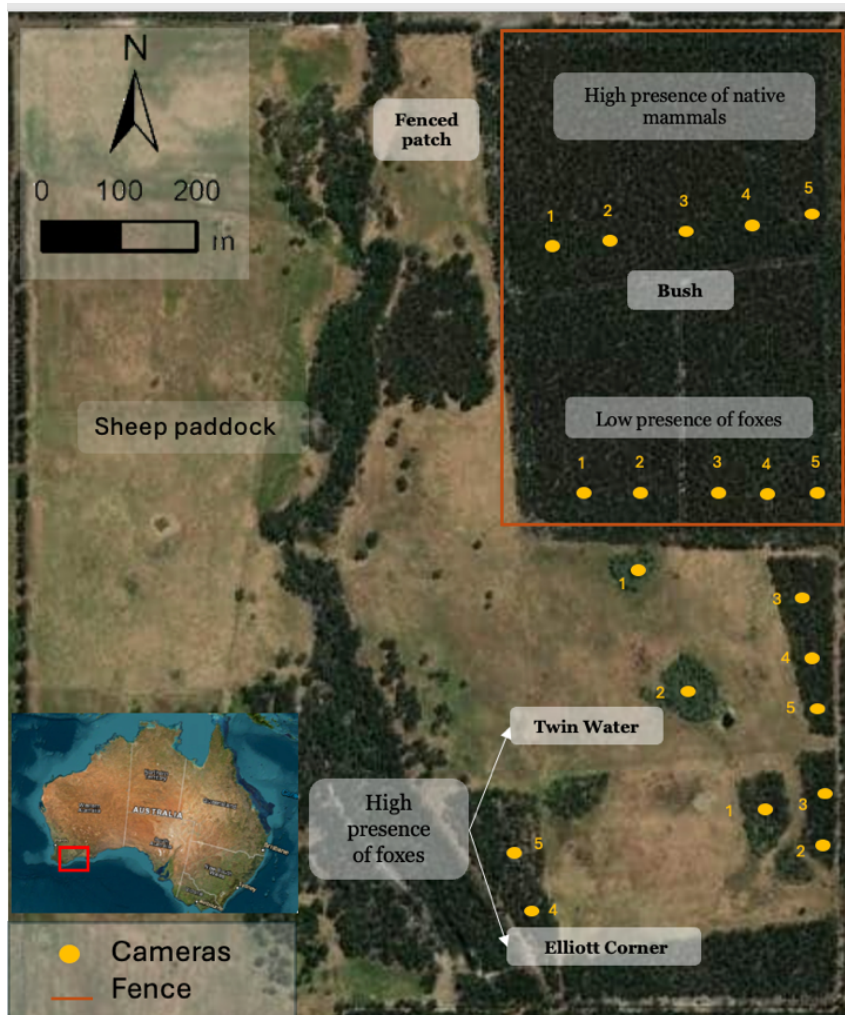
I sincerely thank Dr Barbara Cook and Dr Kelsey Pool for their guidance in experimental design and ongoing academic support throughout the project. I also gratefully acknowledge the owners of the farming property near Narrikup for granting access to conduct this study on their land. Appreciation is extended to all collaborators and fellow ENV4422 students from Semester 1, 2025, for their cooperation in assisting with camera retrieval, vegetation surveys, and species identification from camera trap data.

References

- Fischer J, Lindenmayer DB (2007) Landscape modification and habitat fragmentation: a synthesis. *Global Ecology and Biogeography* 16, 265–280.
- Holland GJ, Bennett AF (2009) Differing responses to landscape change: implications for small mammal assemblages in forest fragments. *Biodiversity and Conservation* 18, 2997–3016.
- Jones ME, Bain GC, Hamer RP, Proft KM, Gardiner RZ, Dixon KJ, Kittipalawattapanol K, Zepeda de Alba AL, Ranyard CE, Munks SA, Barmuta LA, Burridge CP, Johnson CN, Davidson NJ (2021) Research supporting restoration aiming to make a fragmented landscape ‘functional’ for native wildlife. *Ecological Management & Restoration* 22(S2), 65–73.
- Molloy SW, Davis RA (2017) Resilience to agricultural habitat modification in a marsupial folivore. *Australian Mammalogy* 39, 181–191.
- Moore HA, Valentine LE, Dunlop JA, Nimmo DG, Ritchie EG, Hobbs RJ (2021) A rocky heart in a spinifex sea: occurrence of an endangered marsupial predator is multiscale dependent in naturally fragmented landscapes. *Landscape Ecology* 36, 1359–1376.
- Sazima I (2020) Australian Raven (*Corvus coronoides*) scavenges on all five major vertebrate groups at urban Sydney, southeast Australia. *Tropical Natural History* 20, 89–94.
- Short J, Hide A, Stone M (2011) Habitat requirements of the endangered red-tailed phascogale, *Phascogale calura*. *Wildlife Research* 38, 359–369.
- Yeatman GJ, Wayne AF, Mills HR, Prince J (2016) Temporal patterns in the abundance of a critically endangered marsupial relate to disturbance by roads and agriculture. *PLoS ONE* 11(8), e0160790.

Appendix

Appendix 1. Camera trap placement map



Appendix 1. Camera placements (yellow circles) across three remnant patches (Bush, Twin Water, Elliott Corner) on a farm near Narrikup, southwestern Australia. Inset shows study area (ArcGIS Pro basemap). Map created in ArcGIS Pro 3.4; aerial image from Nearmap (2025).